

Networking Basics

For Multiplayer Games Part 1



MCAST

Institute of Information &
Communication Technology

✉ david.deguara@mcast.edu.mt



IP Addressing

Public IP Address

- Assigned to your modem by your Internet Service Provider (ISP).
- Globally unique, allowing communication with external networks.
- Critical for hosting games or connecting to external servers.

The screenshot displays a web interface for checking one's public IP address and location. It includes fields for IPv4 and IPv6 addresses, a table of IP information (ISP, City, Region, Country), a map showing the location, and buttons for hiding the IP address and updating the location.

My IP Address is:

IPv4: ? **46.10.12.107**

IPv6: ? **Not detected**

My IP Information:

ISP:	Go P.L.C.
City:	Sliema
Region:	Sliema
Country:	Malta

Your location may be exposed!

HIDE MY IP ADDRESS NOW

[Show Complete IP Details](#)

Location not accurate?
[Update My IP Location](#)

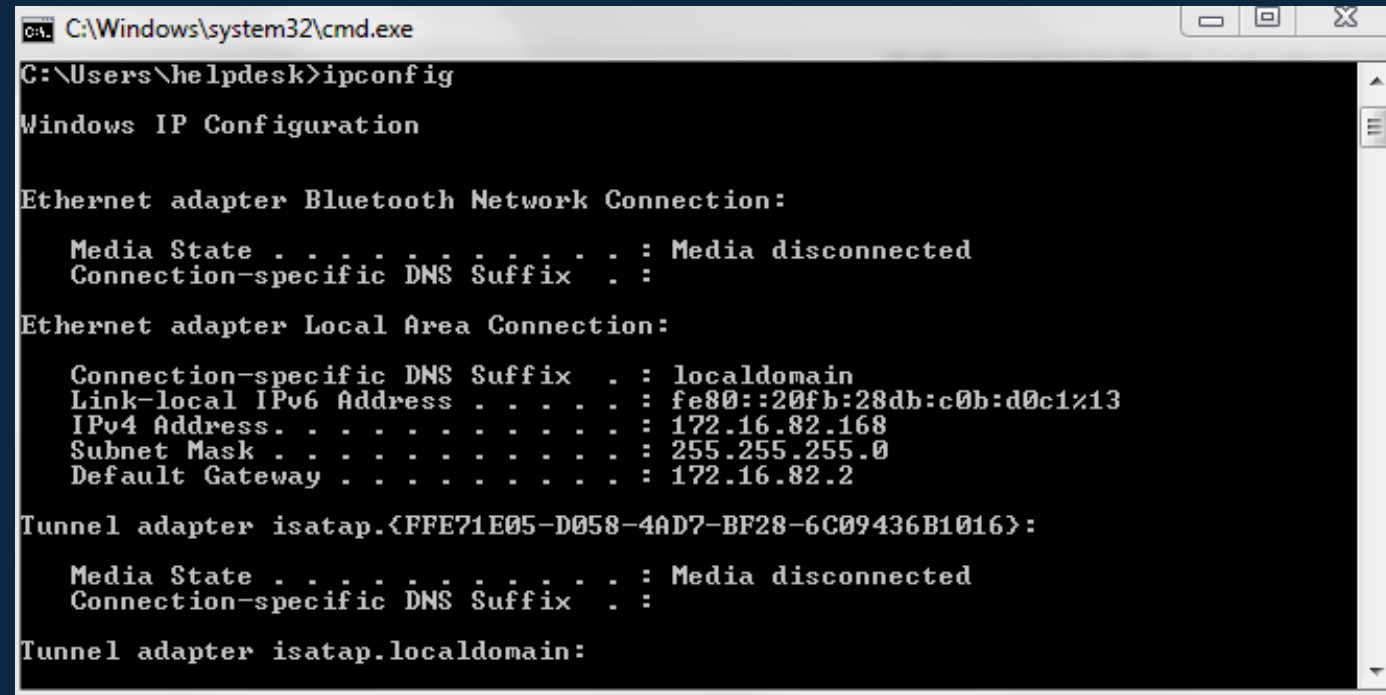
Click for more details about **46.11.73.191**

Malta

Leaflet | © OpenStreetMap Terms

Private IP Address

- Local network address used for communication within homes/offices.
- Hidden from external networks; connects devices like PCs and consoles locally.
- Allows multiple devices to share a single public IP via Network Address Translation (NAT).



```
C:\Windows\system32\cmd.exe
C:\Users\helpdesk>ipconfig

Windows IP Configuration

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::20fb:28db:c0b:d0c1%13
    IPv4 Address. . . . . : 172.16.82.168
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.82.2

Tunnel adapter isatap.{FFE71E05-D058-4AD7-BF28-6C09436B1016}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
Tunnel adapter isatap.localdomain:
```


The background is a vibrant, abstract composition. It features large, overlapping organic shapes in shades of red, orange, purple, and blue. Interspersed among these shapes are patterns of fine, concentric lines and grids of small white dots on darker backgrounds. The overall effect is dynamic and modern.

Understanding Network Ports

What is a Network Port?

- A network port acts like a door number, ensuring data reaches the correct application on a device.
- **Port 80**: Used for HTTP (unsecured web browsing).
- **Port 443**: Used for HTTPS (secure web browsing).
- **Port 21**: Handles FTP (file transfer between computers).
- **Port 25**: Associated with SMTP for email delivery.
- **Port 110**: Used for retrieving emails (POP3).

Network Ports

Why do different applications use specific ports, and what could happen if the same port is used for multiple services?





Port Forwarding in Gaming

What is Port Forwarding?

- Directs data from specific network ports on a public IP to a device on a private network.
- Essential for hosting multiplayer games behind routers or firewalls.
- Used in scenarios like game servers, video chat, and remote access.

How to Set Up Port Forwarding

1. **Identify the Port:** Determine the specific port number used by the game server.
2. **Access Router Settings:** Log in to the router's web interface.
3. **Configure Port Forwarding:** Add a new rule mapping the game server's port to your local device.
4. **Save and Test:** Save changes, restart the router, and test the connection.



Port Forwarding

What are the security risks involved with port forwarding, and how can these risks be mitigated?

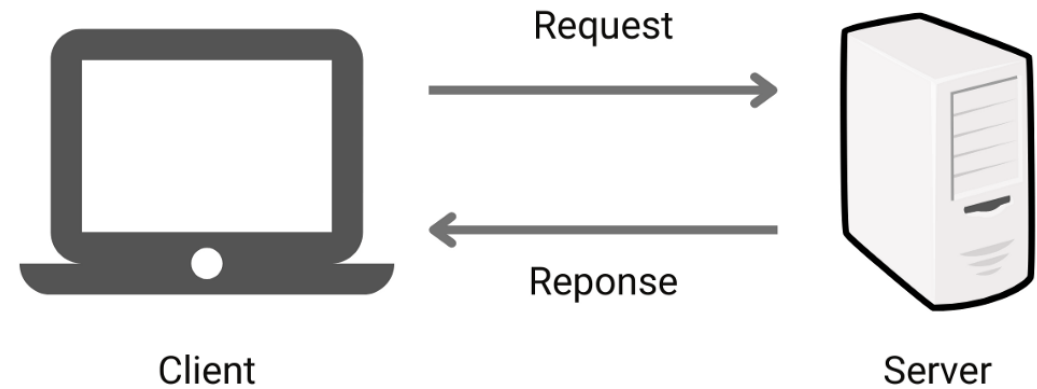




Communication Models

Communication over a network

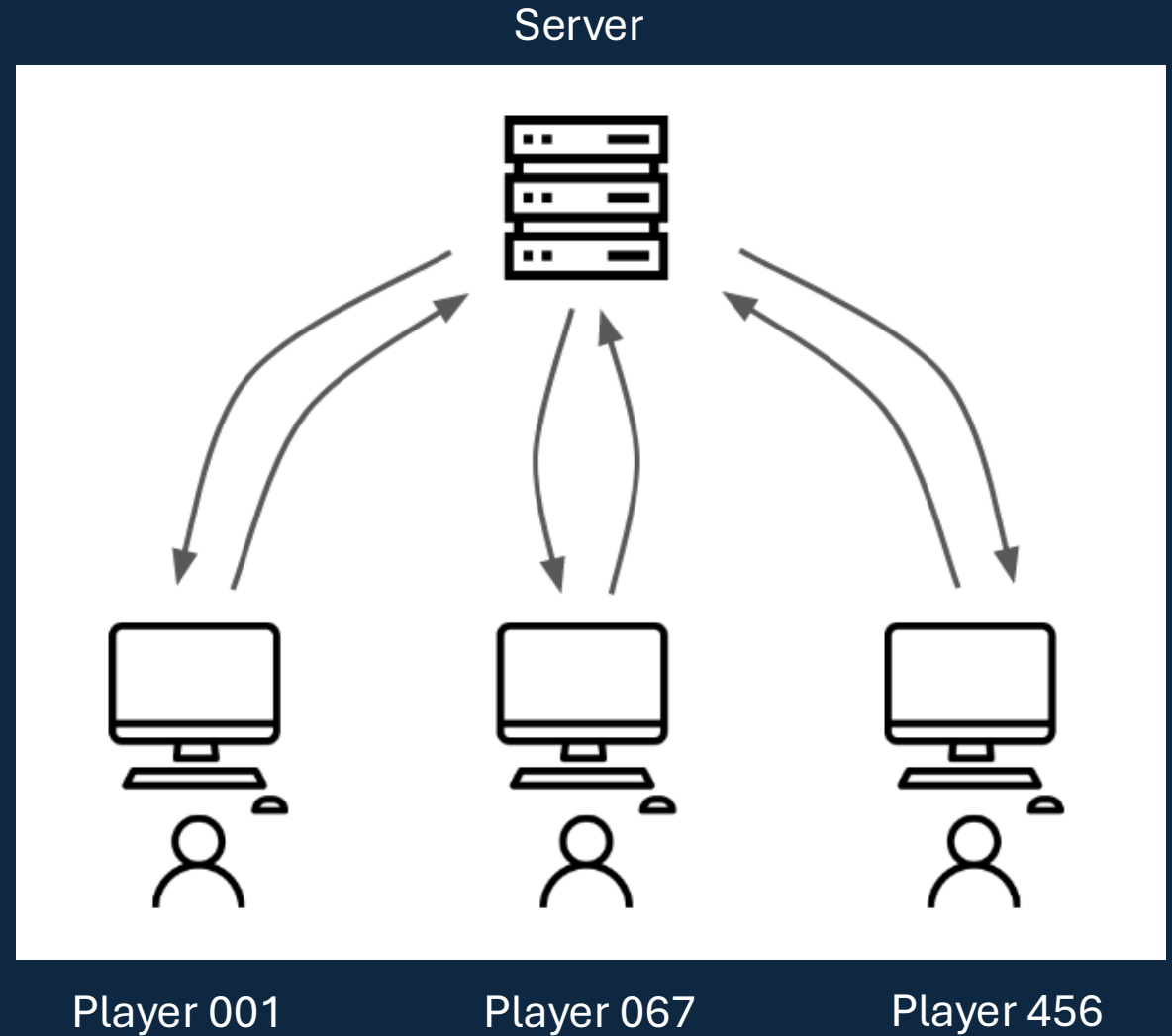
Communication
over a network



Client-Server Model Overview

- **Central Server:** Acts as the authority, managing the game state, rules, and communication.
- **Clients:** Each player's device connects to the server, sending inputs and receiving updates.

Client – Server Model



Advantages of Client-Server Model

- **Security:** Centralised control ensures game rules are enforced.
- **Consistency:** Keeps game state synchronised across all players.
- **Scalability:** Suitable for large-scale games with many players (e.g., MMORPGs).

Client – Server Model

What are the main reasons large-scale games prefer the client-server model over peer-to-peer?



The background is a vibrant, abstract composition. It features large, overlapping organic shapes in shades of red, orange, purple, and blue. Interspersed among these shapes are patterns of fine, concentric white lines and rectangular grids of small white dots. A semi-transparent horizontal band in a muted red color spans the middle of the image, serving as a backdrop for the title text.

The Client's Role

Client Responsibilities

- **Rendering and User Interface (UI):** Displays the game world, controls, and characters.
- **Input Processing:** Captures player inputs and transmits them to the server.
- **Local Processing:** Utilises local resources (CPU, GPU) to reduce server load, but relies on the server for game state.

Network Latency Considerations

- **Impact of Latency:** Higher latency can affect game responsiveness, leading to “lag”.
- **Mitigating Latency:** Servers often reduce latency by placing regional servers close to players.

Network Latency

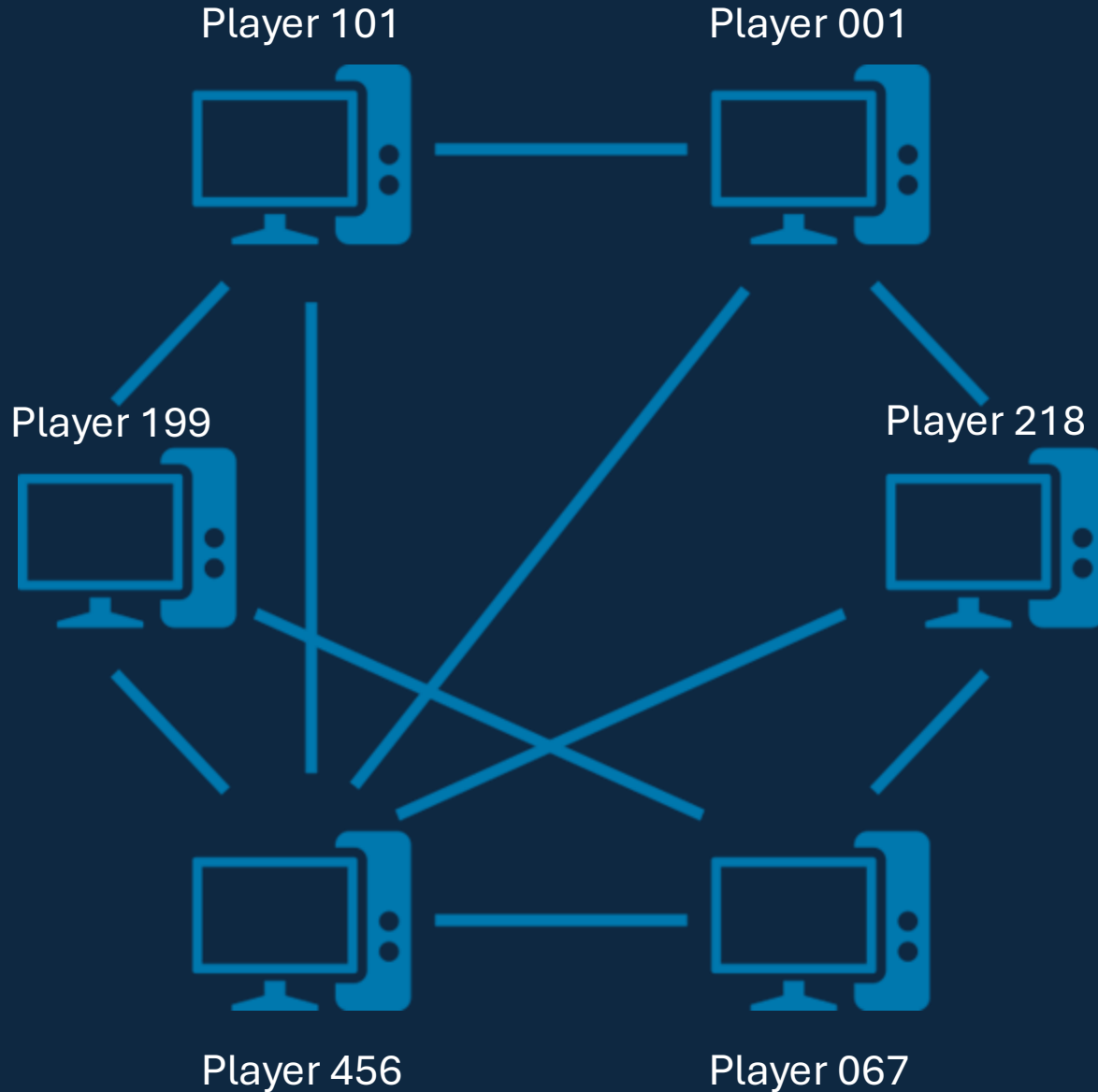
How can network latency impact the fairness and enjoyment of online games?



The background is a vibrant, abstract composition. It features large, flowing organic shapes in shades of magenta, purple, and orange. Overlaid on these are several circular and semi-circular elements, some of which are filled with fine horizontal or vertical lines. A semi-transparent horizontal band in a reddish-pink hue serves as a backdrop for the text.

Peer-to-Peer (P2P)

Peer to Peer (P2P) Model

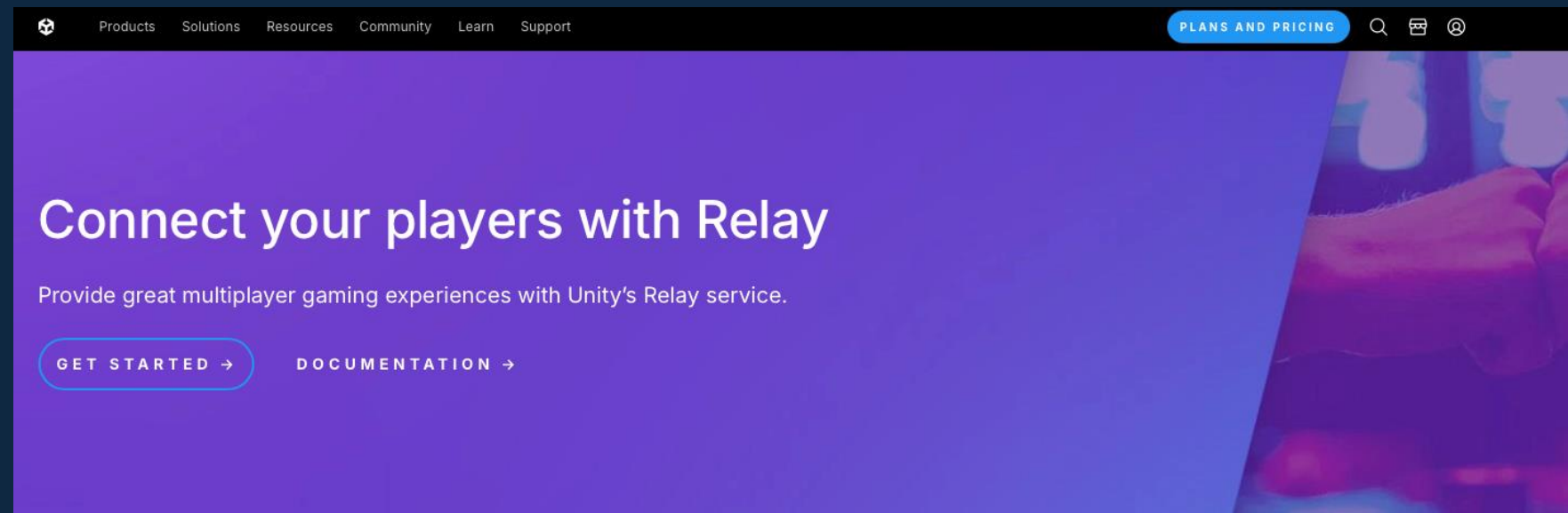


Overview of P2P Model

- **Decentralised:** Each player's device communicates directly with others, without a central server.
- **Use in Early Games:** Frequently used in older multiplayer games (e.g., early *Call of Duty* titles).

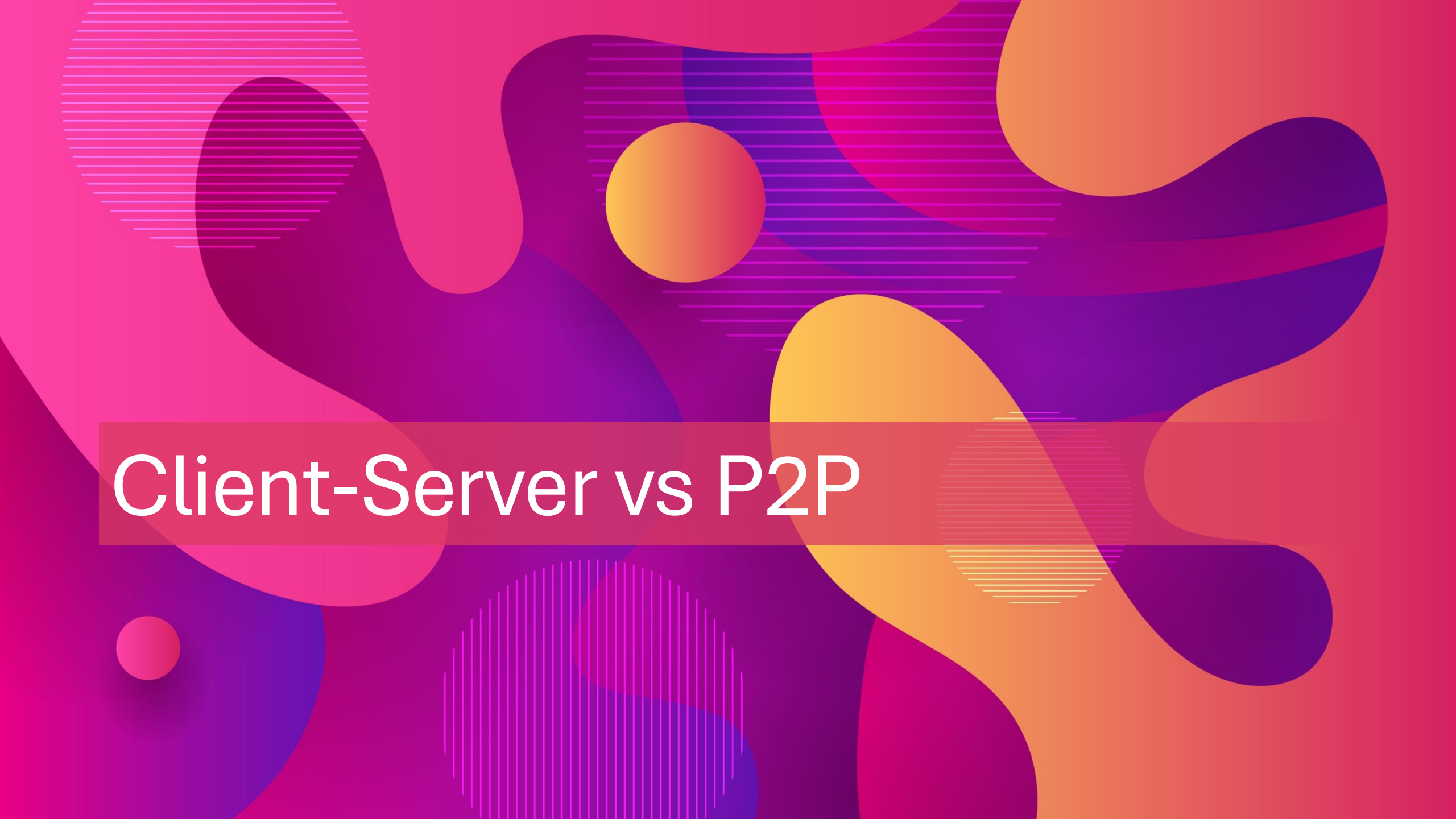
Benefits of P2P Model

- **Lower Costs:** No need for expensive servers.
- **Potential for Lower Latency:** In some cases, direct communication reduces network delays.
- Can be implemented with [Unity Relay](#) to bypass Port-Forwarding.



Challenges of P2P Model

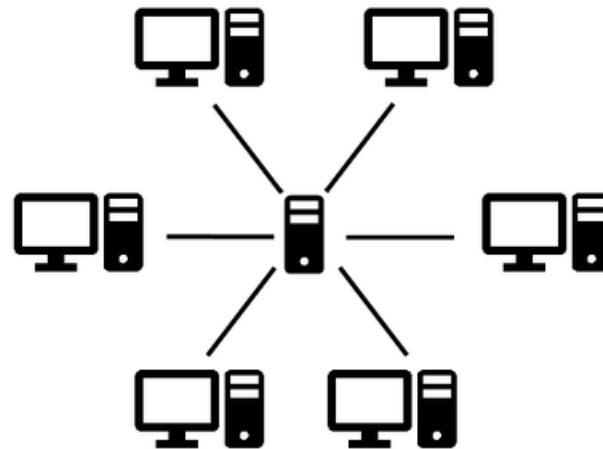
- **Security:** No central authority leads to increased risk of cheating.
- **Synchronisation:** Keeping game states consistent between peers can be difficult.
- **Latency Variability:** Geographic distance between peers can cause lag.



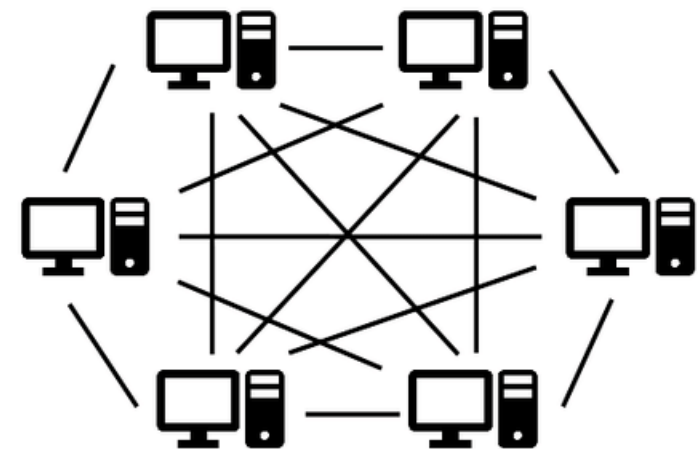
Client-Server vs P2P

Key Differences

- **Client-Server:** Centralised authority ensures security and consistent game states, but requires more resources.
- **P2P:** Decentralised, reducing infrastructure costs but increasing complexity and security risks.



Client-Server network



P2P network

Scalability and Use Cases

- **Client-Server:** Ideal for large, competitive games (e.g., *Fortnite*, *World of Warcraft*).
- **P2P:** Suitable for smaller, informal games with fewer players (e.g., early indie multiplayer games).

What security measures can be implemented to mitigate the risks of cheating in peer-to-peer games?



PLEASE BE MINDFUL OF OTHERS WHO ARE USING THE LAB AFTER YOU



- Plug back in power cables that are unplugged and connect HDMI / keyboard & mouse back to PCs
- Throw away any rubbish you have brought in with you outside
- Put your chair back under the desk and keep everything orderly for the person using it after you
- Report any PCs that are not working to your lecturer

THANK YOU